

Chapter 9

Laplace Transforms

48 Introduction

In recent years there has been a considerable growth of interest in the use of Laplace transforms as an efficient method for solving certain types of differential and integral equations. In addition to such applications, Laplace transforms also have a number of close connections with important parts of pure mathematics. We shall try to give the reader an adequate idea of some of these matters without dwelling too much on the analytic fine points and computational techniques that would be appropriate in a more extensive treatment.

Before entering into the details, we offer a few general remarks aimed at placing the ideas of this chapter in their proper context. We begin by noting that the operation of differentiation transforms a function $f(x)$ into another function, its derivative $f'(x)$. If the letter D is used to denote differentiation, then this transformation can be written

$$D[f(x)] = f'(x). \quad (1)$$

Another important transformation of functions is that of integration:

$$I[f(x)] = \int_0^x f(t) dt. \quad (2)$$

An even simpler transformation is the operation of multiplying all functions by a specific function $g(x)$:

$$M_g[f(x)] = g(x)f(x). \quad (3)$$

The basic feature these examples have in common is that *each transformation operates on functions to produce other functions*. It is clear that in most cases some restriction must be placed on the functions $f(x)$ to which a given

transformation is applied. Thus, in (1) $f(x)$ must be differentiable, and in (2) it must be integrable. In each of our examples, the function on the right is called the *transform* of $f(x)$ under the corresponding transformation.

A general transformation T of functions is said to be *linear* if the relation

$$T[\alpha f(x) + \beta g(x)] = \alpha T[f(x)] + \beta T[g(x)] \quad (4)$$

holds for all admissible functions $f(x)$ and $g(x)$ and all constants α and β . Verbally, equation (4) says that the transform of any linear combination of two functions is the same linear combination of their transforms. It is worth observing that (4) reduces to

$$T[f(x) + g(x)] = T[f(x)] + T[g(x)]$$

and

$$T[\alpha f(x)] = \alpha T[f(x)]$$

when $\alpha = \beta = 1$ and when $\beta = 0$. It is easy to see that the transformations defined by (1), (2), and (3) are all linear.

A class of linear transformations of particular importance is that of the *integral transformations*. To get an idea of what these are, we consider functions $f(x)$ defined on a finite or infinite interval $a \leq x \leq b$, and we choose a fixed function $K(p, x)$ of the variable x and a parameter p . Then the general integral transformation is given by

$$T[f(x)] = \int_a^b K(p, x) f(x) dx = F(p). \quad (5)$$

The function $K(p, x)$ is called the *kernel* of the transformation T , and it is clear that T is linear regardless of the nature of K . The concept of a linear integral transformation, in generalized form, has been the source of some of the most fruitful ideas in modern analysis. Also, in classical analysis, various special cases of (5) have been minutely studied, and have led to specific transformations useful in handling particular types of problems.

When $a = 0$, $b = \infty$, and $K(p, x) = e^{-px}$, we obtain the special case of (5) that concerns us—the *Laplace transformation* L , defined by

$$L[f(x)] = \int_0^{\infty} e^{-px} f(x) dx = F(p). \quad (6)$$

Thus, the Laplace transformation L acts on any function $f(x)$ for which this integral exists, and produces its *Laplace transform* $L[f(x)] = F(p)$, a function of

the parameter p .¹ We remind the reader that the improper integral in (6) is defined to be the following limit, and exists only when this limit exists:

$$\int_0^{\infty} e^{-px} f(x) dx = \lim_{b \rightarrow \infty} \int_0^b e^{-px} f(x) dx. \quad (7)$$

When the limit on the right exists, the improper integral on the left is said to *converge*.

The following Laplace transforms are quite easy to compute:

$$f(x) = 1, \quad F(p) = \int_0^{\infty} e^{-px} dx = \frac{1}{p}; \quad (8)$$

$$f(x) = x, \quad F(p) = \int_0^{\infty} e^{-px} x dx = \frac{1}{p^2}; \quad (9)$$

$$f(x) = x^n, \quad F(p) = \int_0^{\infty} e^{-px} x^n dx = \frac{n!}{p^{n+1}}; \quad (10)$$

$$f(x) = e^{ax}, \quad F(p) = \int_0^{\infty} e^{-px} e^{ax} dx = \frac{1}{p-a}; \quad (11)$$

$$f(x) = \sin ax, \quad F(p) = \int_0^{\infty} e^{-px} \sin ax dx = \frac{a}{p^2 + a^2}; \quad (12)$$

$$f(x) = \cos ax, \quad F(p) = \int_0^{\infty} e^{-px} \cos ax dx = \frac{p}{p^2 + a^2}. \quad (13)$$

The integral in (11) converges for $p > a$, and all the others converge for $p > 0$. Students should perform the necessary calculations themselves, so that the source of these restrictions on p is perfectly clear (see Problem 1). As an illustration, we provide the details for (10), in which n is assumed to be a positive integer:

¹ As this remark suggests, we shall consistently use small letters to denote functions of x and the corresponding capital letters to denote the transforms of these functions.

$$\begin{aligned}
 L[x^n] &= \int_0^{\infty} e^{-px} x^n dx = -\frac{x^n e^{-px}}{p} \Big|_0^{\infty} + \frac{n}{p} \int_0^{\infty} e^{-px} x^{n-1} dx \\
 &= \frac{n}{p} L[x^{n-1}] = \frac{n}{p} \left(\frac{n-1}{p} \right) L[x^{n-2}] \\
 &= \cdots = \frac{n!}{p^n} L[1] = \frac{n!}{p^{n+1}}.
 \end{aligned}$$

It will be noted that we have made essential use here of the fact that

$$\lim_{x \rightarrow \infty} \frac{x^n}{e^{px}} = 0 \quad \text{for } p > 0.$$

The above formulas will be found in [Table 1](#) in [Section 50](#). Additional simple transforms can readily be determined without integration by using the linearity of L , as in

$$L[2x + 3] = 2L[x] + 3L[1] = \frac{2}{p^2} + \frac{3}{p}.$$

In later sections we shall develop methods for finding Laplace transforms of more complicated functions.

As we stated above, the Laplace transformation L can be regarded as the special case of the general integral transformation (5) obtained by taking $a = 0$, $b = \infty$, and $K(p, x) = e^{-px}$. Why do we choose these limits and this particular kernel? In order to see why this might be a fruitful choice, it is useful to consider a suggestive analogy with power series.

If we write a power series in the form

$$\sum_{n=0}^{\infty} a(n)x^n,$$

then its natural analog is the improper integral

$$\int_0^{\infty} a(t)x^t dt.$$

We now change the notation slightly by writing $x = e^{-p}$, and this integral becomes

$$\int_0^{\infty} e^{-pt} a(t) dt,$$

which is precisely the Laplace transform of the function $a(t)$. Laplace transforms are therefore the continuous analogs of power series; and since power series are important in analysis, we have reasonable grounds for expecting that Laplace transforms will also be important.

A short account of Laplace is given in [Appendix A](#).

Problems

1. Evaluate the integrals in (8), (9), (11), (12), and (13).
2. Without integrating, show that
 - (a) $L[\sinh ax] = \frac{a}{p^2 - a^2}, \quad p > |a|;$
 - (b) $L[\cosh ax] = \frac{p}{p^2 - a^2}, \quad p > |a|.$
3. Find $L[\sin^2 ax]$ and $L[\cos^2 ax]$ without integrating. How are these two transforms related to one another?
4. Use the formulas given in the text to find the transform of each of the following functions:
 - (a) $10;$
 - (b) $x^5 + \cos 2x;$
 - (c) $2e^{3x} - \sin 5x;$
 - (d) $4 \sin x \cos x + 2e^{-x};$
 - (e) $x^6 \sin^2 3x + x^6 \cos^2 3x.$
5. Find a function $f(x)$ whose transform is
 - (a) $\frac{30}{p^4};$
 - (b) $\frac{2}{p+3};$
 - (c) $\frac{4}{p^3} + \frac{6}{p^2+4};$
 - (d) $\frac{1}{p^2+p};$
 - (e) $\frac{1}{p^4+p^2}.$
6. Give a reasonable definition of $\frac{1}{2}!$.

49 A Few Remarks on the Theory

Before proceeding to the applications, it is desirable to consider more carefully the circumstances under which a function has a Laplace transform. A detailed and rigorous treatment of this problem would require familiarity with the general theory of improper integrals, which we do not assume. On the other hand, it is customary to give a brief introduction to this subject in elementary calculus, and a grasp of the following simple statements will suffice for our purposes.

First, the integral

$$\int_0^{\infty} f(x) dx \quad (1)$$

is said to *converge* if the limit

$$\lim_{b \rightarrow \infty} \int_0^b f(x) dx$$

exists, and in this case the value of (1) is by definition the value of this limit:

$$\int_0^{\infty} f(x) dx = \lim_{b \rightarrow \infty} \int_0^b f(x) dx.$$

Next, (1) converges whenever the integral

$$\int_0^{\infty} |f(x)| dx$$

converges, and in this case (1) is said to *converge absolutely*. And finally, (1) converges absolutely—and therefore converges—if there exists a function $g(x)$ such that $|f(x)| \leq g(x)$ and

$$\int_0^{\infty} g(x) dx$$

converges (this is known as the *comparison test*).

Accordingly, if $f(x)$ is a given function defined for $x \geq 0$, the convergence of (1) requires first of all that the integral $\int_0^b f(x) dx$ must exist for each finite $b > 0$. To guarantee this, it suffices to assume that $f(x)$ is continuous, or at least is *piecewise continuous*. By the latter we mean that $f(x)$ is continuous over

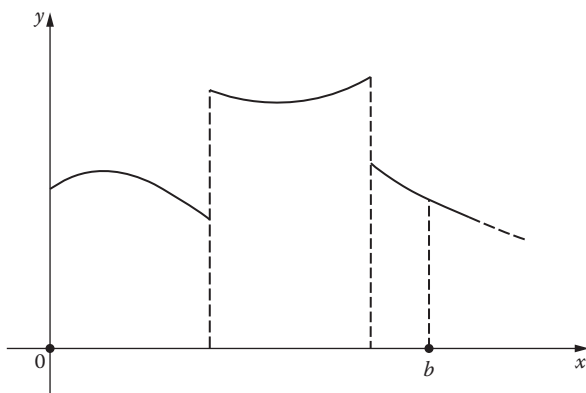


FIGURE 64

every finite interval $0 \leq x \leq b$, except possibly at a finite number of points where there are jump discontinuities, at which the function approaches different limits from the left and right. Figure 64 illustrates the appearance of a typical piecewise continuous function; its integral from 0 to b is the sum of the integrals of its continuous parts over the corresponding subintervals. This class of functions contains virtually all that are likely to arise in practice. In particular, it includes the discontinuous step functions and sawtooth functions expressing the sudden application or removal of forces and voltages in problems of physics and engineering.

If $f(x)$ is piecewise continuous for $x \geq 0$, then the only remaining threat to the existence of its Laplace transform

$$F(p) = \int_0^{\infty} e^{-px} f(x) dx$$

is the behavior of the integrand $e^{-px}f(x)$ for large x . In order to make sure that this integrand diminishes rapidly enough for convergence—or that $f(x)$ does not grow too rapidly—we shall further assume that $f(x)$ is of *exponential order*. This means that there exist constants M and c such that

$$|f(x)| \leq Me^{cx}. \quad (2)$$

Thus, although $f(x)$ may become infinitely large as $x \rightarrow \infty$, it must grow less rapidly than a multiple of some exponential function e^{cx} . It is clear that any bounded function is of exponential order with $c = 0$. As further examples, we mention e^{ax} (with $c = a$) and x^n (with c any positive number). On the other hand, e^{x^2} is not of exponential order. If $f(x)$ satisfies (2), then we have

$$|e^{-px}f(x)| \leq Me^{-(p-c)x},$$

and since the integral of the function on the right converges for $p > c$, the Laplace transform of $f(x)$ converges absolutely for $p > c$. In addition, we note that

$$\begin{aligned} |F(p)| &= \left| \int_0^{\infty} e^{-px} f(x) dx \right| \leq \int_0^{\infty} |e^{-px} f(x)| dx \\ &\leq M \int_0^{\infty} e^{-(p-c)x} dx = \frac{M}{p-c}, \quad p > c, \end{aligned}$$

so

$$F(p) \rightarrow 0 \quad \text{as } p \rightarrow \infty. \quad (3)$$

Actually, it can be shown that (3) is true whenever $F(p)$ exists, regardless of whether or not $f(x)$ is piecewise continuous and of exponential order. Thus, if $\phi(p)$ is a function of p with the property that its limit as $p \rightarrow \infty$ does not exist or is not equal to zero, then it cannot be the Laplace transform of *any* $f(x)$. In particular, polynomials in p , $\sin p$, $\cos p$, e^p , and $\log p$ cannot be Laplace transforms. On the other hand, a rational function is a Laplace transform if the degree of the numerator is less than that of the denominator.

The above remarks show that any piecewise continuous function of exponential order has a Laplace transform, so these conditions are sufficient for the existence of $L[f(x)]$. However, they are not necessary, as the example $f(x) = x^{-1/2}$ shows. This function has an infinite discontinuity at $x = 0$, so it is not piecewise continuous, but nevertheless its integral from 0 to b exists; and since it is bounded for large x , its Laplace transform exists. Indeed, for $p > 0$ we have

$$L[x^{-1/2}] = \int_0^{\infty} e^{-px} x^{-1/2} dx,$$

and the change of variable $px = t$ gives

$$L[x^{-1/2}] = p^{-1/2} \int_0^{\infty} e^{-t} t^{-1/2} dt.$$

Another change of variable, $t = s^2$, leads to

$$L[x^{-1/2}] = 2p^{-1/2} \int_0^{\infty} e^{-s^2} ds. \quad (4)$$

In most treatments of elementary calculus it is shown that the last-written integral has the value $\sqrt{\pi}/2$ (see Problem 1), so we have

$$L[x^{-1/2}] = \sqrt{\frac{\pi}{p}}. \quad (5)$$

This result will be useful in a later section.

In the remainder of this chapter we shall concentrate on the uses of Laplace transforms, and will not attempt to study the purely mathematical theory behind our procedures. Naturally these procedures need justification, and readers who are impatient with formalism can find what they want in more extensive discussions of the subject.

Problems

1. If f denotes the integral in (4), then (s being a dummy variable) we can write

$$I^2 = \left(\int_0^\infty e^{-x^2} dx \right) \left(\int_0^\infty e^{-y^2} dy \right) = \iint_{0 \leq x, y < \infty} e^{-(x^2+y^2)} dx dy.$$

Evaluate this double integral by changing to polar coordinates, and thereby show that $I = \sqrt{\pi}/2$.

2. In each of the following cases, graph the function and find its Laplace transform:
 - (a) $f(x) = u(x - a)$ where a is a positive number and $u(x)$ is the unit step function defined by

$$u(x) = \begin{cases} 0 & \text{if } x < 0 \\ 1 & \text{if } x \geq 0; \end{cases}$$

- (b) $f(x) = [x]$ where $[x]$ denotes the greatest integer $\leq x$;
 - (c) $f(x) = x - [x]$;
 - (d) $f(x) = \begin{cases} \sin x & \text{if } 0 \leq x \leq \pi \\ 0 & \text{if } x > \pi. \end{cases}$
3. Show explicitly that $L[e^{x^2}]$ does not exist. *Hint:* $x^2 - px = (x - p/2)^2 - p^2/4$.
 4. Show explicitly that $L[x^{-1}]$ does not exist.

5. Let ϵ be a positive number and consider the function $f_\epsilon(x)$ defined by

$$f_\epsilon(x) = \begin{cases} 1/\epsilon & \text{if } 0 \leq x \leq \epsilon \\ 0 & \text{if } x > \epsilon. \end{cases}$$

The graph of this function is shown in [Figure 65](#). It is clear that for every $\epsilon > 0$ we have $\int_0^\infty f_\epsilon(x) dx = 1$. Show that

$$L[f_\epsilon(x)] = \frac{1 - e^{-p\epsilon}}{p\epsilon}$$

and

$$\lim_{\epsilon \rightarrow \infty} L[f_\epsilon(x)] = 1.$$

Strictly speaking, $\lim_{\epsilon \rightarrow 0} f_\epsilon(x)$ does not exist as a function, so $L[\lim_{\epsilon \rightarrow 0} f_\epsilon(x)]$ is not defined; but if we throw caution to the winds, then

$$\delta(x) = \lim_{\epsilon \rightarrow 0} f_\epsilon(x)$$

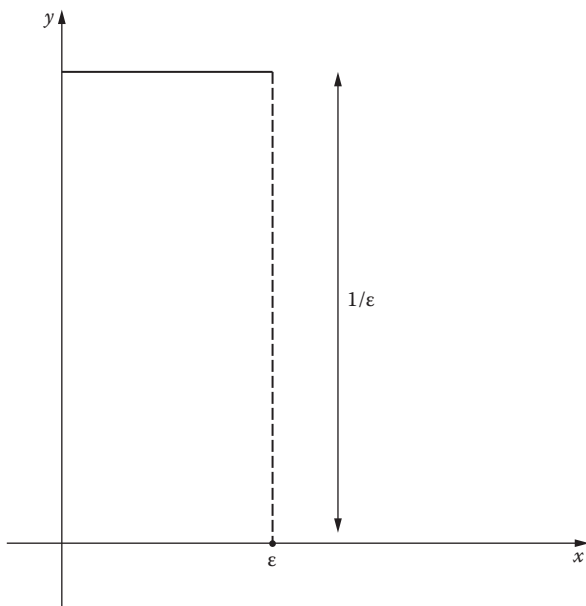


FIGURE 65

is seen to be some kind of quasi-function that is infinite at $x = 0$ and zero for $x > 0$, and has the properties

$$\int_0^{\infty} \delta(x) dx = 1 \quad \text{and} \quad L[\delta(x)] = 1.$$

This quasi-function $\delta(x)$ is called the *Dirac delta function* or *unit impulse function*.²

50 Applications to Differential Equations

Suppose we wish to find the particular solution of the differential equation

$$y'' + ay' + by = f(x) \tag{1}$$

that satisfies the initial conditions $y(0) = y_0$ and $y'(0) = y'_0$. It is clear that we could try to apply the methods of [Chapter 3](#) to find the general solution and then evaluate the arbitrary constants in accordance with the given initial conditions. However, the use of Laplace transforms provides an alternate way of attacking this problem that has several advantages.

To see how this method works, let us apply the Laplace transformation L to both sides of (1):

$$L[y'' + ay' + by] = L[f(x)].$$

By the linearity of L , this can be written as

$$L[y''] + aL[y'] + bL[y] = L[f(x)]. \tag{2}$$

Our next step is to express $L[y']$ and $L[y'']$ in terms of $L[y]$. First, an integration by parts gives

² P.A.M. Dirac (1902–1984) was an English theoretical physicist who won the Nobel Prize at the age of thirty-one for his work in quantum theory. There are several ways of making good mathematical sense out of his delta function. See, for example, I. Halperin, *Introduction to the Theory of Distributions*, University of Toronto Press, Toronto, 1952; or A. Erdélyi, *Operational Calculus and Generalized Functions*, Holt, New York, 1962. Dirac's own discussion of his function is interesting and easy to read; see pp. 58–61 of his treatise *The Principles of Quantum Mechanics*, Oxford University Press, 4th ed., 1958.

$$\begin{aligned}
 L[y'] &= \int_0^{\infty} e^{-px} y' dx \\
 &= ye^{-px} \Big|_0^{\infty} + p \int_0^{\infty} e^{-px} y dx \\
 &= -y(0) + pL[y],
 \end{aligned}$$

so

$$L[y'] = pL[y] - y(0). \quad (3)$$

Next,

$$L[y''] = L[(y')'] = pL[y'] - y'(0),$$

so

$$L[y''] = p^2L[y] - py(0) - y'(0). \quad (4)$$

If we now insert the given initial conditions in (3) and (4), and substitute these expressions in (2), we obtain an algebraic equation for $L[y]$,

$$p^2L[y] - py_0 - y'_0 + apL[y] - ay_0 + bL[y] = L[f(x)];$$

and solving for $L[y]$ yields

$$L[y] = \frac{L[f(x)] + (p+a)y_0 + y'_0}{p^2 + ap + b}. \quad (5)$$

The function $f(x)$ is known, so its Laplace transform $L[f(x)]$ is a specific function of p ; and since a , b , y_0 , and y'_0 are known constants, $L[y]$ is completely known as a function of p . If we can now find which function $y(x)$ has the right side of equation (5) as its Laplace transform, then this function will be the solution of our problem—initial conditions and all. These procedures are particularly suited to solving equations of the form (1) in which the function $f(x)$ is discontinuous, for in this case the methods of [Chapter 3](#) may be difficult to apply.

There is an obvious flaw in this discussion: in order for (2) to have any meaning, the functions $f(x)$, y , y' , and y'' must have Laplace transforms. This difficulty can be remedied by simply assuming that $f(x)$ is piecewise continuous and of exponential order. Once this assumption is made, then it can be shown (we omit the proof) that y , y' , and y'' necessarily have the same

properties, so they also have Laplace transforms. Another difficulty is that in obtaining (3) and (4) we took it for granted that

$$\lim_{x \rightarrow \infty} y e^{-px} = 0 \quad \text{and} \quad \lim_{x \rightarrow \infty} y' e^{-px} = 0.$$

However, since y and y' are automatically of exponential order, these statements are valid for all sufficiently large values of p .

Example 1. Find the solution of

$$y'' + 4y = 4x \tag{6}$$

that satisfies the initial conditions $y(0) = 1$ and $y'(0) = 5$.

When L is applied to both sides of (6), we get

$$L[y''] + 4L[y] = 4L[x]. \tag{7}$$

If we recall that $L(x) = 1/p^2$, and use (4) and the initial conditions, then (7) becomes

$$p^2 L[y] - p - 5 + 4L[y] = \frac{4}{p^2}$$

or

$$(p^2 + 4)L[y] = p + 5 + \frac{4}{p^2},$$

so

$$\begin{aligned} L[y] &= \frac{p}{p^2 + 4} + \frac{5}{p^2 + 4} + \frac{4}{p^2(p^2 + 4)} \\ &= \frac{p}{p^2 + 4} + \frac{5}{p^2 + 4} + \frac{1}{p^2} - \frac{1}{p^2 + 4} \\ &= \frac{p}{p^2 + 4} + \frac{4}{p^2 + 4} + \frac{1}{p^2}. \end{aligned} \tag{8}$$

On referring to the transforms obtained in [Section 48](#), we see that (8) can be written

$$\begin{aligned} L[y] &= L[\cos 2x] + L[2 \sin 2x] + L[x] \\ &= L[\cos 2x + 2 \sin 2x + x], \end{aligned}$$

so

$$y = \cos 2x + 2 \sin 2x + x$$

is the desired solution. We can easily check this result, for the general solution of (6) is seen by inspection to be

$$y = c_1 \cos 2x + c_2 \sin 2x + x,$$

and the initial conditions imply at once that $c_1 = 1$ and $c_2 = 2$.

The validity of this procedure clearly rests on the assumption that only one function $y(x)$ has the right side of equation (8) as its Laplace transform. This is true if we restrict ourselves to continuous $y(x)$'s—and any solution of a differential equation is necessarily continuous. When $f(x)$ is assumed to be continuous, the equation $L[f(x)] = F(p)$ is often written in the form

$$L^{-1}[F(p)] = f(x).$$

It is customary to call L^{-1} the *inverse Laplace transformation*, and to refer to $f(x)$ as the *inverse Laplace transform* of $F(p)$. Since L is linear, it is evident that L^{-1} is also linear. In Example 1 we made use of the following inverse transforms:

$$L^{-1}\left[\frac{p}{p^2 + 4}\right] = \cos 2x, \quad L^{-1}\left[\frac{2}{p^2 + 4}\right] = \sin 2x, \quad L^{-1}\left[\frac{1}{p^2}\right] = x.$$

This example also illustrates the value of decomposition into partial fractions as a method of finding inverse transforms.

For the convenience of the reader, we give a short list of useful transform pairs in Table 1. Much more extensive tables are available for the use of those who find it desirable to apply Laplace transforms frequently in their work.

We shall consider a number of general properties of Laplace transforms that greatly increase the flexibility of Table 1. The first of these is the *shifting formula*:

$$L[e^{ax}f(x)] = F(p - a). \quad (9)$$

To establish this, it suffices to observe that

$$\begin{aligned} L[e^{ax}f(x)] &= \int_0^{\infty} e^{-px} e^{ax} f(x) dx \\ &= \int_0^{\infty} e^{-(p-a)x} f(x) dx \\ &= F(p - a). \end{aligned}$$

Formula (9) can be used to find transforms of products of the form $e^{ax}f(x)$ when $F(p)$ is known, and also to find inverse transforms of functions of the form $F(p - a)$ when $f(x)$ is known.

TABLE 1

Simple Transform Pairs

$f(x)$	$F(p) = L[f(x)]$
1	$\frac{1}{p}$
x	$\frac{1}{p^2}$
x^n	$\frac{n!}{p^{n+1}}$
e^{ax}	$\frac{1}{p-a}$
$\sin ax$	$\frac{a}{p^2 + a^2}$
$\cos ax$	$\frac{p}{p^2 + a^2}$
$\sinh ax$	$\frac{a}{p^2 - a^2}$
$\cosh ax$	$\frac{p}{p^2 - a^2}$

Example 2.

$$L[\sin bx] = \frac{b}{p^2 + b^2},$$

so

$$L[e^{ax} \sin bx] = \frac{b}{(p-a)^2 + b^2}.$$

Example 3.

$$L^{-1}\left[\frac{1}{p^2}\right] = x,$$

so

$$L^{-1}\left[\frac{1}{(p-a)^2}\right] = e^{ax}x.$$

The methods of this section can be applied to systems of linear differential equations with constant coefficients, and also to certain types of partial differential equations. Discussions of these further applications can be found in more extended works on Laplace transforms.³

³ For example, see R. V. Churchill, *Operational Mathematics*, 2d ed., McGraw-Hill, New York, 1958.

Problems

- Find the Laplace transforms of
 - $x^5 e^{-2x}$;
 - $(1-x^2)e^{-x}$;
 - $e^{3x} \cos 2x$.
- Find the inverse Laplace transforms of
 - $\frac{6}{(p+2)^2+9}$;
 - $\frac{12}{(p+3)^4}$;
 - $\frac{p+3}{p^2+2p+5}$.
- Solve each of the following differential equations by the method of Laplace transforms:
 - $y' + y = 3e^{2x}$, $y(0)=0$;
 - $y'' - 4y' + 4y = 0$, $y(0) = 0$ and $y'(0) = 3$;
 - $y'' + 2y' + 2y = 2$, $y(0) = 0$ and $y'(0) = 1$;
 - $y'' + y' = 3x^2$, $y(0) = 0$ and $y'(0) = 1$;
 - $y'' + 2y' + 5y = 3e^{-x} \sin x$, $y(0) = 0$ and $y'(0) = 3$.
- Find the solution of $y'' - 2ay' + a^2y = 0$ in which the initial conditions $y(0) = y_0$ and $y'(0) = y'_0$ are left unrestricted. (This provides an additional derivation of our earlier solution, in [Section 17](#), for the case in which the auxiliary equation has a double root.)
- Apply (3) to establish the formula for the Laplace transform of an integral,

$$L \left[\int_0^x f(x) dx \right] = \frac{F(p)}{p},$$

and verify this by finding

$$L^{-1} \left[\frac{1}{p(p+1)} \right]$$

in two ways.

- Solve $y' + 4y + 5 \int_0^x y dx = e^{-x}$, $y(0) = 0$.

51 Derivatives and Integrals of Laplace Transforms

Consider the general Laplace transform formula

$$F(p) = \int_0^{\infty} e^{-px} f(x) dx.$$

The differentiation of this with respect to p under the integral sign can be justified, and yields

$$F'(p) = \int_0^{\infty} e^{-px} (-x) f(x) dx \quad (1)$$

or

$$L[-xf(x)] = F'(p). \quad (2)$$

By differentiating (1), we find that

$$L[x^2 f(x)] = F''(p), \quad (3)$$

and, more generally, that

$$L[(-1)^n x^n f(x)] = F^{(n)}(p) \quad (4)$$

for any positive integer n . These formulas can be used to find transforms of functions of the form $x^n f(x)$ when $F(p)$ is known.

Example 1. Since $L[\sin ax] = a/(p^2 + a^2)$, we have

$$L[x \sin ax] = -\frac{d}{dp} \left(\frac{a}{p^2 + a^2} \right) = \frac{2ap}{(p^2 + a^2)^2}.$$

Example 2. We know from [Section 49](#) that $L[x^{-1/2}] = \sqrt{\pi/p}$, so

$$L[x^{1/2}] = L[x(x^{-1/2})] = -\frac{d}{dp} \left(\sqrt{\frac{\pi}{p}} \right) = \frac{1}{2p} \sqrt{\frac{\pi}{p}}.$$

If we apply (2) to a function $y(x)$ and its derivatives—and remember formulas 50-(3) and 50-(4)—then we get

$$L[xy] = -\frac{d}{dp} L[y] = -\frac{dY}{dp}, \quad (5)$$

$$L[xy'] = -\frac{d}{dp}L[y'] = -\frac{d}{dp}[pY - y(0)] = -\frac{d}{dp}[pY], \quad (6)$$

and

$$\begin{aligned} L[xy''] &= -\frac{d}{dp}L[y''] = -\frac{d}{dp}[p^2Y - py(0) - y'(0)] \\ &= -\frac{d}{dp}[p^2Y - py(0)]. \end{aligned} \quad (7)$$

These formulas can sometimes be used to solve linear differential equations whose coefficients are first degree polynomials in the independent variable.

Example 3. Bessel's equation of order zero is

$$xy'' + y' + xy = 0. \quad (8)$$

It is known to have a single solution $y(x)$ with the property that $y(0) = 1$. To find this solution, we apply L to (8) and use (5) and (7), which gives

$$-\frac{d}{dp}[p^2Y - p] + pY - 1 - \frac{dY}{dp} = 0$$

or

$$(p^2 + 1)\frac{dY}{dp} = -pY. \quad (9)$$

If we separate the variables in (9) and integrate, we get

$$\begin{aligned} Y &= \frac{c}{\sqrt{p^2 + 1}} = c(p^2 + 1)^{-1/2} \\ &= \frac{c}{p} \left(1 + \frac{1}{p^2} \right)^{-1/2}. \end{aligned} \quad (10)$$

On expanding the last factor by the binomial series

$$\begin{aligned} (1+z)^a &= 1 + az + \frac{a(a-1)}{2!}z^2 + \frac{a(a-1)(a-2)}{3!}z^3 + \dots \\ &\quad + \frac{a(a-1)\cdots(a-n+1)}{n!}z^n + \dots, \end{aligned}$$

(10) becomes

$$\begin{aligned}
 Y &= \frac{c}{p} \left[1 - \frac{1}{2} \cdot \frac{1}{p^2} + \frac{1}{2!} \cdot \frac{1}{2} \cdot \frac{3}{2} \cdot \frac{1}{p^4} - \frac{1}{3!} \cdot \frac{1}{2} \cdot \frac{3}{2} \cdot \frac{5}{2} \cdot \frac{1}{p^6} + \dots \right. \\
 &\quad \left. + \frac{1 \cdot 3 \cdot 5 \dots (2n-1)}{2^n n!} \frac{(-1)^n}{p^{2n}} + \dots \right] \\
 &= c \sum_{n=0}^{\infty} \frac{(2n)!}{2^{2n} (n!)^2} \frac{(-1)^n}{p^{2n+1}}.
 \end{aligned}$$

If we now proceed formally, and compute the inverse transform of this series term by term, then we find that

$$\begin{aligned}
 y(x) &= c \sum_{n=0}^{\infty} \frac{(-1)^n}{2^{2n} (n!)^2} x^{2n} \\
 &= c \left(1 - \frac{x^2}{2^2} + \frac{x^4}{2^2 \cdot 4^2} - \frac{x^6}{2^2 \cdot 4^2 \cdot 6^2} + \dots \right).
 \end{aligned}$$

Since $y(0) = 1$, it follows that $c = 1$, and our solution is

$$y(x) = 1 - \frac{x^2}{2^2} + \frac{x^4}{2^2 \cdot 4^2} - \frac{x^6}{2^2 \cdot 4^2 \cdot 6^2} + \dots$$

This series defines the important Bessel function $J_0(x)$, whose Laplace transform we have found to be $1/\sqrt{p^2+1}$. We obtained this series in [Chapter 8](#) in a totally different way, and it is interesting to see how easily it can be derived by Laplace transform methods.

We now turn to the problem of integrating transforms, and our main result is

$$L \left[\frac{f(x)}{x} \right] = \int_p^{\infty} F(p) dp. \quad (11)$$

To establish this, we put $L[f(x)/x] = G(p)$. An application of (2) yields

$$\frac{dG}{dp} = L \left[(-x) \frac{f(x)}{x} \right] = -L[f(x)] = -F(p),$$

so

$$G(p) = - \int_a^p F(p) dp$$

for some a . Since we want to make $G(p) \rightarrow 0$ as $p \rightarrow \infty$, we put $a = \infty$ and get

$$G(p) = \int_p^{\infty} F(p) dp,$$

which is (11). This formula is useful in finding transforms of functions of the form $f(x)/x$ when $F(p)$ is known. Furthermore, if we write (11) as

$$\int_0^{\infty} e^{-px} \frac{f(x)}{x} dx = \int_p^{\infty} F(p) dp$$

and let $p \rightarrow 0$, we obtain

$$\int_0^{\infty} \frac{f(x)}{x} dx = \int_0^{\infty} F(p) dp, \quad (12)$$

which is valid whenever the integral on the left exists. This formula can sometimes be used to evaluate integrals that are difficult to handle by other methods.

Example 4. Since $L[\sin x] = 1/(p^2 + 1)$, (12) gives

$$\int_0^{\infty} \frac{\sin x}{x} dx = \int_0^{\infty} \frac{dp}{p^2 + 1} = \tan^{-1} p \Big|_0^{\infty} = \frac{\pi}{2}.$$

For easy reference, we list the main general properties of Laplace transforms in Table 2. It will be noted that the last item in the list is new. We shall discuss this formula and its applications in the next section.

TABLE 2

General Properties of $L[f(x)] = F(p)$

$$L[\alpha f(x) + \beta g(x)] = \alpha F(p) + \beta G(p)$$

$$L[e^{ax}f(x)] = F(p - a)$$

$$L[f'(x)] = pF(p) - f(0);$$

$$L[p^n f(x)] = p^n F(p) - pf(0) - f'(0)$$

$$L\left[\int_0^x f(x) dx\right] = \frac{F(p)}{p}$$

$$L[-xf(x)] = F'(p);$$

$$L[(-1)^n x^n f(x)] = F^{(n)}(p)$$

$$L\left[\frac{f(x)}{x}\right] = \int_p^{\infty} F(p) dp$$

$$L\left[\int_0^x f(x-t)g(t) dt\right] = F(p)G(p)$$

Problems

1. Show that

$$L[x \cos ax] = \frac{p^2 - a^2}{(p^2 + a^2)^2},$$

and use this result to find

$$L^{-1} \left[\frac{1}{(p^2 + a^2)^2} \right].$$

2. Find each of the following transforms:

- (a) $L[x^2 \sin ax]$;
 (b) $L[x^{3/2}]$.

3. Solve each of the following differential equations:

- (a) $xy'' + (3x - 1)y' - (4x + 9)y = 0$, $y(0) = 0$;
 (b) $xy'' + (2x + 3)y' + (x + 3)y = 3e^{-x}$, $y(0) = 0$.

4. If $y(x)$ satisfies the differential equation

$$y'' + x^2y = 0,$$

where $y(0) = y_0$ and $y'(0) = y'_0$, show that its transform $Y(p)$ satisfies the equation

$$Y'' + p^2Y = py_0 + y'_0.$$

Observe that the second equation is of the same type as the first, so that no progress has been made. The method of Example 3 is advantageous only when the coefficients are first degree polynomials.

5. If a and b are positive constants, evaluate the following integrals:

- (a) $\int_0^\infty \frac{e^{-ax} - e^{-bx}}{x} dx$;
 (b) $\int_0^\infty \frac{e^{-ax} \sin bx}{x} dx$.

6. Show formally that

- (a) $\int_0^\infty J_0(x) dx = 1$;
 (b) $J_0(x) = \frac{1}{\pi} \int_0^\pi \cos(x \cos t) dt$.

7. If $x > 0$, show formally that

$$(a) \quad f(x) = \int_0^\infty \frac{\sin xt}{t} dt = \frac{\pi}{2};$$

$$(b) \quad f(x) = \int_0^\infty \frac{\cos xt}{1+t^2} dt = \frac{\pi}{2} e^{-x}.$$

8. (a) If $f(x)$ is periodic with period a , so that $f(x+a) = f(x)$, show that

$$F(p) = \frac{1}{1-e^{-ap}} \int_0^a e^{-px} f(x) dx.$$

(b) Find $F(p)$ if $f(x) = 1$ in the intervals from 0 to 1, 2 to 3, 4 to 5, etc., and $f(x) = 0$ in the remaining intervals.

52 Convolutions and Abel's Mechanical Problem

If $L[f(x)] = F(p)$ and $L[g(x)] = G(p)$, what is the inverse transform of $F(p)G(p)$?

To answer this question formally, we use dummy variables s and t in the integrals defining the transforms and write

$$\begin{aligned} F(p)G(p) &= \left[\int_0^\infty e^{-ps} f(s) ds \right] \left[\int_0^\infty e^{-pt} g(t) dt \right] \\ &= \int_0^\infty \int_0^\infty e^{-p(s+t)} f(s) g(t) ds dt \\ &= \int_0^\infty \left[\int_0^\infty e^{-p(s+t)} f(s) ds \right] g(t) dt, \end{aligned}$$

where the integration is extended over the first quadrant ($s \geq 0, t \geq 0$) in the st -plane. We now introduce a new variable x in the inner integral of the last expression by putting $s+t = x$, so that $s = x-t$ and (t being fixed during this integration) $ds = dx$. This enables us to write

$$\begin{aligned} F(p)G(p) &= \int_0^\infty \left[\int_t^\infty e^{-px} f(x-t) dx \right] g(t) dt \\ &= \int_0^\infty \int_t^\infty e^{-px} f(x-t) g(t) dx dt. \end{aligned}$$

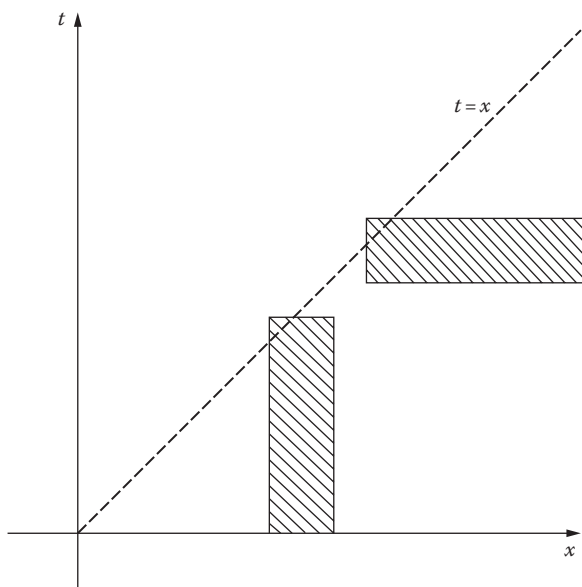


FIGURE 66

This integration is extended over the first half of the first quadrant ($x-t \geq 0$) in the xt -plane, and reversing the order as suggested in [Figure 66](#), we get

$$\begin{aligned}
 F(p)G(p) &= \int_0^{\infty} \left[\int_0^x e^{-px} f(x-t)g(t) dt \right] dx \\
 &= \int_0^{\infty} e^{-px} \left[\int_0^x f(x-t)g(t) dt \right] dx \\
 &= L \left[\int_0^x f(x-t)g(t) dt \right]. \tag{1}
 \end{aligned}$$

The integral in the last expression is a function of the upper limit x , and provides the answer to our question. This integral is called the *convolution* of the functions $f(x)$ and $g(x)$. It can be regarded as a “generalized product” of these functions. The fact stated in equation (1)—namely, that the product of the Laplace transforms of two functions is the transform of their convolution—is called the *convolution theorem*.

The convolution theorem can be used to find inverse transforms. For instance, since $L[x] = 1/p^2$ and $L[\sin x] = 1/(p^2 + 1)$, we have

$$\begin{aligned}
 L^{-1}\left[\frac{1}{p^2(p^2+1)}\right] &= L^{-1}\left[\frac{1}{p^2}\left(\frac{1}{p^2+1}\right)\right] \\
 &= \int_0^x (x-t) \sin t \, dt \\
 &= x - \sin x,
 \end{aligned}$$

as can easily be verified by partial fractions. A more interesting class of applications arises as follows. If $f(x)$ and $k(x)$ are given functions, then the equation

$$f(x) = y(x) + \int_0^x k(x-t)y(t) \, dt, \quad (2)$$

in which the unknown function $y(x)$ appears under the integral sign, is called an *integral equation*. Because of its special form, in which the integral is the convolution of the two functions $k(x)$ and $y(x)$, this equation lends itself to solution by means of Laplace transforms. In fact, if we apply L to both sides of equation (2), we get

$$L[f(x)] = L[y(x)] + L[k(x)]L[y(x)],$$

so

$$L[y(x)] = \frac{L[f(x)]}{1 + L[k(x)]}. \quad (3)$$

The right side of (3) is presumably known as a function of p ; and if this function is a recognizable transform, then we have our solution $y(x)$.

Example 1. The integral equation

$$y(x) = x^3 + \int_0^x \sin(x-t)y(t) \, dt \quad (4)$$

is of this type, and by applying L we get

$$L[y(x)] = L[x^3] + L[\sin x]L[y(x)].$$

Solving for $L[y(x)]$ yields

$$\begin{aligned}
 L[y(x)] &= \frac{L[x^3]}{1 - L[\sin x]} = \frac{3!/p^4}{1 - 1/(p^2 + 1)} \\
 &= \frac{3!}{p^4} \left(\frac{p^2 + 1}{p^2} \right) = \frac{3!}{p^4} + \frac{3!}{p^6},
 \end{aligned}$$

so

$$y(x) = x^3 + \frac{1}{20}x^5$$

is the solution of (4).

As a further illustration of this technique, we analyze a classical problem in mechanics that leads to an integral equation of the above type. Consider a wire bent into a smooth curve (Figure 67) and let a bead of mass m start from rest and slide without friction down the wire to the origin under the action of its own weight. Suppose that (x, y) is the starting point and (u, v) is any intermediate point. If the shape of the wire is specified by a given function $y = y(x)$, then the total time of descent will be a definite function $T(y)$ of the initial height y . *Abel's mechanical problem* is the converse: specify the function $T(y)$ in advance and then find the shape of the wire that yields this $T(y)$ as the total time of descent.

To formulate this problem mathematically, we start with the principle of conservation of energy:

$$\frac{1}{2}m\left(\frac{ds}{dt}\right)^2 = mg(y-v) \quad \text{or} \quad -\frac{ds}{dt} = \sqrt{2g(y-v)},$$

which can be written as

$$dt = -\frac{ds}{\sqrt{2g(y-v)}}.$$

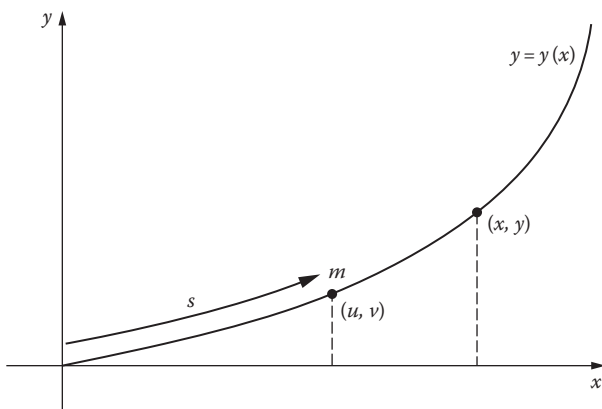


FIGURE 67

On integrating this from $v = y$ to $v = 0$, we get

$$T(y) = \int_{v=y}^{v=0} dt = \int_{v=0}^{v=y} \frac{ds}{\sqrt{2g(y-v)}} = \frac{1}{\sqrt{2g}} \int_0^y \frac{s'(v)dv}{\sqrt{y-v}}. \quad (5)$$

Now

$$s = s(y) = \int_0^y \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$$

is known whenever the curve $y = y(x)$ is known, so its derivative

$$f(y) = s'(y) = \sqrt{1 + \left(\frac{dx}{dy}\right)^2} \quad (6)$$

is also known. If we insert (6) in (5), then we see that

$$T(y) = \frac{1}{\sqrt{2g}} \int_0^y \frac{f(v)dv}{\sqrt{y-v}}, \quad (7)$$

and this enables us to calculate $T(y)$ whenever the curve is given. In Abel's problem we want to find the curve when $T(y)$ is given; and from this point of view, the function $f(y)$ in equation (7) is the unknown and (7) itself is called *Abel's integral equation*. Note that the integral in (7) is the convolution of the functions $y^{-1/2}$ and $f(y)$, so on applying the Laplace transformation L we get

$$L[T(y)] = \frac{1}{\sqrt{2g}} L[y^{-1/2}]L[f(y)].$$

If we now recall that $L[y^{-1/2}] = \sqrt{\pi/p}$, then this yields

$$\begin{aligned} L[f(y)] &= \sqrt{2g} \frac{L[T(y)]}{\sqrt{\pi/p}} \\ &= \sqrt{\frac{2g}{\pi}} p^{1/2} L[T(y)]. \end{aligned} \quad (8)$$

When $T(y)$ is given, the right side of equation (8) is known as a function of p , so hopefully we can find $f(y)$ by taking the inverse transform. Once $f(y)$ is known, the curve itself can be found by solving the differential equation (6).

As a concrete example, we now specialize our discussion to the case in which $T(y)$ is a constant T_0 . This assumption means that the time of descent is to be independent of the starting point. The curve defined by this property is called the *tautochrone*, so our problem is that of finding the tautochrone. In this case, (8) becomes

$$L[f(y)] = \sqrt{\frac{2g}{\pi}} p^{1/2} L[T_0] = \sqrt{\frac{2g}{\pi}} p^{1/2} \frac{T_0}{p} = b^{1/2} \sqrt{\frac{\pi}{p}},$$

where $b = 2g T_0^2 / \pi^2$. The inverse transform of $\sqrt{\pi/p}$ is $y^{-1/2}$, so

$$f(y) = \sqrt{\frac{b}{y}}. \quad (9)$$

With this $f(y)$, (6) now yields

$$1 + \left(\frac{dx}{dy} \right)^2 = \frac{b}{y}$$

as the differential equation of the curve, so

$$x = \int \sqrt{\frac{b-y}{y}} dy.$$

On substituting $y = b \sin^2 \phi$, this becomes

$$\begin{aligned} x &= 2b \int \cos^2 \phi d\phi = b \int (1 + \cos 2\phi) d\phi \\ &= \frac{b}{2} (2\phi + \sin 2\phi) + c, \end{aligned}$$

so

$$x = \frac{b}{2} (2\phi + \sin 2\phi) + c \quad \text{and} \quad y = \frac{b}{2} (1 - \cos 2\phi). \quad (10)$$

The curve must pass through the origin (0,0), so $c = 0$; and if we put $a = b/2$ and $\theta = 2\phi$, then (10) take the simpler form

$$x = a(\theta + \sin \theta) \quad \text{and} \quad y = a(1 - \cos \theta).$$

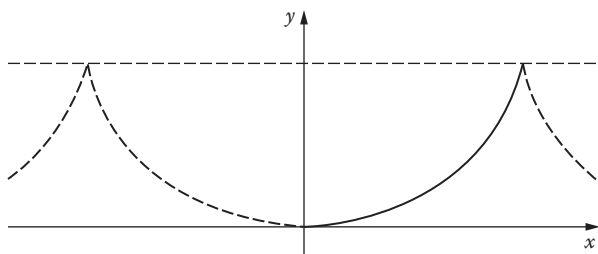


FIGURE 68

These are the parametric equations of the cycloid shown in Figure 68, which is generated by a fixed point on a circle of radius a rolling under the horizontal dashed line $y = 2a$. Since $2a = b = 2g T_0^2 / \pi^2$, the diameter of the generating circle is determined by the constant time of descent.

Accordingly, the tautochrone is a cycloid. In Problems 6–5 and 11–5 we verified this property of cycloids by other methods. Our present discussion has the advantage of enabling us to find the tautochrone without knowing in advance what the answer will be.

Problems

- Find $L^{-1}[1/(p^2 + a^2)^2]$ by convolution. (See Problem 51–1.)
- Solve each of the following integral equations:
 - $y(x) = 1 - \int_0^x (x-t)y(t)dt;$
 - $y(x) = e^x \left[1 + \int_0^x e^{-t}y(t)dt \right];$
 - $e^{-x} = y(x) + 2 \int_0^x \cos(x-t)y(t)dt;$
 - $3 \sin 2x = y(x) + \int_0^x (x-t)y(t)dt.$
- Deduce

$$f(y) = \frac{\sqrt{2g}}{\pi} \frac{d}{dy} \int_0^y \frac{T(t)dt}{\sqrt{y-t}}$$

from equation (8), and use this to verify (9) when $T(y)$ is a constant T_0 .

- Find the equation of the curve of descent if $T(y) = k\sqrt{y}$ for some constant k .

5. Show that the differential equation

$$y'' + a^2y = f(x), \quad y(0) = y'(0) = 0,$$

has

$$y(x) = \frac{1}{a} \int_0^x f(t) \sin a(x-t) dt$$

as its solution.

53 More about Convolutions. The Unit Step and Impulse Functions

In the preceding section we found that the product of the Laplace transforms of two functions is the transform of a certain combination of these functions called their *convolution*. If we use the time t as the independent variable and if the two functions are $f(t)$ and $g(t)$, then this *convolution theorem* [equation 52-(1)] can be expressed as follows:

$$L[f(t)]L[g(t)] = L\left[\int_0^t f(t-\tau)g(\tau)d\tau\right]. \quad (1)$$

It is customary to denote the convolution of $f(t)$ and $g(t)$ by $f(t)*g(t)$, so that

$$f(t)*g(t) = \int_0^t f(t-\tau)g(\tau)d\tau. \quad (2)$$

The convolution theorem (1) can then be written in the form

$$L[f(t)*g(t)] = L[f(t)]L[g(t)]. \quad (3)$$

Our purpose in this section is to discuss an application of this theorem that makes it possible to determine the response of a mechanical or electrical system to a general stimulus if its response to the unit step function is known. These ideas have important uses in electrical engineering and other areas of applied science.

Any physical system capable of responding to a stimulus can be thought of as a device that transforms an *input function* (the stimulus) into an *output*

function (the response). If we assume that all initial conditions are zero at the moment $t = 0$ when the input $f(t)$ begins to act, then by setting up the differential equation that describes the system, operating on this equation with the Laplace transformation L , and solving for the transform of the output $y(t)$, we obtain an equation of the form

$$L[y(t)] = \frac{L[f(t)]}{z(p)}, \quad (4)$$

where $z(p)$ is a polynomial whose coefficients depend only on the parameters of the system itself. This equation is the main source of the explicit formulas for $y(t)$ that we obtain below with the aid of the convolution theorem.

Let us be more specific. We seek solutions $y(t)$ of the linear differential equation

$$y'' + ay' + by = f(t) \quad (5)$$

that satisfy the initial conditions

$$y(0) = y'(0) = 0 \quad (6)$$

describing a mechanical or electrical system at rest in its equilibrium position. The input $f(t)$ can be thought of as an impressed external force F or electromotive force E that begins to act at time $t = 0$, as discussed in [Section 20](#). When this input is the unit step function $u(t)$ defined in Problem 49–2(a), the solution (or output) $y(t)$ is denoted by $A(t)$ and called the *indicial response*; that is,

$$A'' + aA' + bA = u(t).$$

By applying the Laplace transformation L and using formulas (3) and (4) in [Section 50](#), we obtain

$$p^2 L[A] + apL[A] + bL[A] = L[u(t)] = \frac{1}{p},$$

so

$$L[A] = \frac{1}{p} \frac{1}{p^2 + ap + b} = \frac{1}{p} \frac{1}{z(p)}, \quad (7)$$

where $z(p)$ is defined by the last equality. We now apply L in the same way to the general equation (5), which yields (4); and dividing both sides of this by p and using (7) gives

$$\frac{1}{p} L[y] = \frac{1}{pz(p)} L[f] = L[A]L[f]. \quad (8)$$

The convolution theorem now enables us to write (8) in the form

$$\frac{1}{p} L[y] = L[A(t) * f(t)] = L\left[\int_0^t A(t-\tau)f(\tau)d\tau\right].$$

By using formula 50-(3) once more we get

$$\begin{aligned} L[y] &= pL\left[\int_0^t A(t-\tau)f(\tau)d\tau\right] \\ &= L\left[\frac{d}{dt}\int_0^t A(t-\tau)f(\tau)d\tau\right], \end{aligned}$$

so

$$y(t) = \frac{d}{dt}\int_0^t A(t-\tau)f(\tau)d\tau. \quad (9)$$

By applying Leibniz's rule for differentiating integrals⁴ to (9), we now have

$$y(t) = \int_0^t A'(t-\tau)f(\tau)d\tau + A(0)f(t). \quad (10)$$

Next, since $L[A]L[f] = L[f]L[A]$, (8) also enables us to write

$$\frac{1}{p} L[y] = L[f(t) * A(t)] = L\left[\int_0^t f(t-\sigma)A(\sigma)d\sigma\right],$$

⁴ Leibniz's rule states that if $F(t) = \int_u^v G(t, x)dx$, where u and v are functions of t and x is a dummy variable, then

$$\frac{d}{dt}F(t) = \int_u^v \frac{\partial}{\partial t}G(t, x)dx + G(t, v)\frac{dv}{dt} - G(t, u)\frac{du}{dt}.$$

and by following the same reasoning as before, we obtain

$$y(t) = \int_0^t f'(t-\sigma)A(\sigma)d\sigma + f(0)A(t). \quad (11)$$

In formula (10) we notice that $A(0) = 0$ because of the initial conditions (6); and (11) takes a more convenient form under the change of variable $\tau = t - \sigma$. Our two formulas (10) and (11) for $y(t)$ therefore become

$$y(t) = \int_0^t A'(t-\tau)f(\tau)d\tau \quad (12)$$

and

$$y(t) = \int_0^t A(t-\tau)f'(\tau)d\tau + f(0)A(t). \quad (13)$$

Each of these formulas provides a solution of (5) for a general input $f(t)$ in terms of the indicial response $A(t)$ to the unit step function. Formula (13) is sometimes called the *principle of superposition*; it has been variously attributed to the famous nineteenth century physicists James Clerk Maxwell and Ludwig Boltzmann, and also to the English applied mathematician Oliver Heaviside.

Example 1. Use formula (13) to solve $y'' + y' - 6y = 2e^{3t}$, where $y(0) = y'(0) = 0$. Here we have

$$L[A(t)] = \frac{1}{p(p^2 + p - 6)},$$

so by partial fractions and inversion we find that

$$A(t) = -\frac{1}{6} + \frac{1}{15}e^{-3t} + \frac{1}{10}e^{2t}.$$

Since $f(t) = 2e^{3t}$, $f'(t) = 6e^{3t}$ and $f(0) = 2$, (13) gives

$$\begin{aligned} y(t) &= \int_0^t \left[-\frac{1}{6} + \frac{1}{15}e^{-3(t-\tau)} + \frac{1}{10}e^{2(t-\tau)} \right] 6e^{3\tau}d\tau \\ &\quad + 2 \left[-\frac{1}{6} + \frac{1}{15}e^{-3t} + \frac{1}{10}e^{2t} \right] \\ &= \frac{1}{3}e^{3t} + \frac{1}{15}e^{-3t} - \frac{2}{5}e^{2t}. \end{aligned}$$

This solution can be verified by substituting directly in the given equation, and also by solving the equation by the method already studied in [Section 50](#).

We can also use formula (12) to solve the equation in this example, but before doing this, it is desirable to express (12) in a simpler form. We accomplish this by using the unit impulse function $\delta(t)$ described in Problem 49–5. In physics, the *impulse* due to a constant force F acting over a time interval Δt is defined to be $F \Delta t$. The “function” $\delta(t)$ can be thought of as a limit of constant functions of unit impulse acting over shorter and shorter intervals of time; it is used to describe forces and voltages that act very suddenly, as in the case of a hammer blow on a mechanical system or a lightning stroke on a transmission line.

For us, the essential property of $\delta(t)$ is that expressed by the equation

$$L[\delta(t)] = 1,$$

obtained in Problem 49–5. When the input $f(t)$ in the differential equation (5) is the unit impulse function $\delta(t)$, the output $y(t)$ is denoted by $h(t)$ and called the *impulsive response*. Applying L in this case yields

$$L[h(t)] = \frac{1}{z(p)}, \quad (14)$$

so

$$h(t) = L^{-1} \left[\frac{1}{z(p)} \right].$$

By (7) and (14),

$$L[A(t)] = \frac{1}{p} \frac{1}{z(p)} = \frac{L[h(t)]}{p},$$

and it follows from Problem 50–5 that

$$A(t) = \int_0^t h(t-\tau) d\tau.$$

This shows that $A'(t) = h(t)$, so formula (12) becomes

$$y(t) = \int_0^t h(t-\tau) f(\tau) d\tau. \quad (15)$$

Thus, the solution of (5) with a general input $f(t)$ can be written as the convolution of the impulsive response $h(t)$ with $f(t)$.

Example 2. Consider again the equation $y'' + y' - 6y = 2e^{3t}$ solved in Example 1. We have

$$h(t) = L^{-1} \left[\frac{1}{(p+3)(p-2)} \right] = \frac{1}{5}(e^{2t} - e^{-3t}),$$

so that

$$\begin{aligned} y(t) &= \int_0^t \frac{1}{5} [e^{2(t-\tau)} - e^{-3(t-\tau)}] 2e^{3\tau} d\tau \\ &= \frac{1}{3}e^{3t} + \frac{1}{15}e^{-3t} - \frac{2}{5}e^{2t}, \end{aligned}$$

as before.

Remark 1. In complicated practical situations electrical engineers are sometimes compelled to work with indicial or impulsive responses $A(t)$ or $h(t)$ that are only accessible experimentally, by means of oscilloscope pictures responding to generator-produced step functions or impulse functions. In such a case the output must be calculated from (13) or (15) by methods of graphical integration that permit the plotting of individual points on the output curve. For a discussion of these topics see [Chapter 9](#) of W. D. Day, *Introduction to Laplace Transforms for Radio and Electronic Engineers*, Interscience, New York, 1960.

Remark 2. To form a more general view of the meaning of convolution let us consider a linear physical system in which the effect at the present time t of a small stimulus $g(\tau) d\tau$ at any past time τ is proportional to the size of the stimulus. We further assume that the proportionality factor depends only on the elapsed time $t - \tau$, and thus has the form $f(t - \tau)$. The effect at the present time t is therefore

$$f(t - \tau)g(\tau) d\tau.$$

Since the system is linear, the total effect at the present time t due to the stimulus acting throughout the entire past history of the system is obtained by adding these separate effects, and this leads to the convolution integral

$$\int_0^t f(t - \tau)g(\tau) d\tau.$$

The lower limit here is 0 because we assume that the stimulus started acting at time $t = 0$, that is, that $g(\tau) = 0$ for $\tau < 0$. The importance of convolution is difficult to exaggerate: it provides a reasonable way of taking account of the past in the study of wave motion, heat conduction, diffusion, and other areas of mathematical physics.

Problems

1. Show that $f(t) * g(t) = g(t) * f(t)$ directly from the definition (2), by introducing a new dummy variable $\sigma = t - \tau$. This shows that the operation of forming convolutions is commutative. It is also associative and distributive:

$$f(t) * [g(t) * h(t)] = [f(t) * g(t)] * h(t)$$

and

$$f(t) * [g(t) + h(t)] = f(t) * g(t) + f(t) * h(t),$$

$$[f(t) + g(t)] * h(t) = f(t) * h(t) + g(t) * h(t).$$

An interesting discussion of the abstract properties of convolution is given by Mark Kac and Stanislaw Ulam on pp. 140–142 of *Mathematics and Logic*, New American Library, New York, 1969.

2. Find the convolution of each of the following pairs of functions:
 - (a) $1, \sin at$;
 - (b) e^{at}, e^{bt} , where $a \neq b$;
 - (c) t, e^{at} ;
 - (d) $\sin at, \sin bt$, where $a \neq b$.
3. Verify the convolution theorem for each of the pairs of functions considered in Problem 2.
4. Use the methods of both Examples 1 and 2 to solve each of the following differential equations:
 - (a) $y'' + 5y' + 6y = 5e^{3t}, y(0) = y'(0) = 0$;
 - (b) $y'' + y' - 6y = t, y(0) = y'(0) = 0$;
 - (c) $y'' - y' = t^2, y(0) = y'(0) = 0$.

5. When the polynomial $z(p)$ has distinct real zeros a and b , so that

$$\frac{1}{z(p)} = \frac{1}{(p-a)(p-b)} = \frac{A}{p-a} + \frac{B}{p-b}$$

for suitable constants A and B , then

$$h(t) = Ae^{at} + Be^{bt}$$

and (15) takes the form

$$y(t) = \int_0^t f(\tau) [Ae^{a(t-\tau)} + Be^{b(t-\tau)}] d\tau.$$

This is sometimes called the *Heaviside expansion theorem*.

- (a) Use this theorem to write the solution of $y'' + 3y' + 2y = f(t)$, $y(0) = y'(0) = 0$.
 - (b) Give an explicit evaluation of the solution in (a) for the cases $f(t) = e^{3t}$ and $f(t) = t$.
 - (c) Find the solutions in (b) by using the superposition principle (13).
6. Formula (13) can also be derived from (4) as follows, without the use of Leibniz's rule for differentiating integrals:

$$\begin{aligned}
 L[y(t)] &= \frac{L[f(t)]}{z(p)} = \frac{1}{pz(p)} \cdot pL[f(t)] \\
 &= L[A(t)] \cdot pL[f(t)] \\
 &= L[A(t)] \cdot \{L[f'(t)] + f(0)\} \\
 &= L[A(t) * f'(t)] + f(0)L[A(t)] \\
 &= L \left[\int_0^t A(t-\tau) f'(\tau) d\tau + f(0)A(t) \right].
 \end{aligned}$$

Check the steps.

7. As we know from [Section 20](#), the forced vibrations of an undamped spring-mass system are described by the differential equation

$$Mx'' + kx = f(t),$$

where $x(t)$ is the displacement and $f(t)$ is the impressed external force or "forcing function." If $x(0) = x'(0) = 0$, find the functions $A(t)$ and $h(t)$ and write down the solution $x(t)$ for any $f(t)$.

8. The current $I(t)$ in an electric circuit with inductance L and resistance R is given by equation (4) in [Section 13](#):

$$L \frac{dI}{dt} + RI = E(t),$$

where $E(t)$ is the impressed electromotive force. If $I(0) = 0$, use the methods of this section to find $I(t)$ in each of the following cases:

- (a) $E(t) = E_0 u(t)$;
- (b) $E(t) = E_0 \delta(t)$;
- (c) $E(t) = E_0 \sin \omega t$.

Appendix A. Laplace

Pierre Simon de Laplace (1749–1827) was a French mathematician and theoretical astronomer who was so famous in his own time that he was known as the Newton of France. His main interests throughout his life were celestial mechanics, the theory of probability, and personal advancement.

At the age of twenty-four he was already deeply engaged in the detailed application of Newton's law of gravitation to the solar system as a whole, in which the planets and their satellites are not governed by the sun alone but interact with one another in a bewildering variety of ways. Even Newton had been of the opinion that divine intervention would occasionally be needed to prevent this complex mechanism from degenerating into chaos. Laplace decided to seek reassurance elsewhere, and succeeded in proving that the ideal solar system of mathematics is a stable dynamical system that will endure unchanged for all time. This achievement was only one of the long series of triumphs recorded in his monumental treatise *Mécanique Céleste* (published in five volumes from 1799 to 1825), which summed up the work on gravitation of several generations of illustrious mathematicians. Unfortunately for his later reputation, he omitted all reference to the discoveries of his predecessors and contemporaries, and left it to be inferred that the ideas were entirely his own. Many anecdotes are associated with this work. One of the best known describes the occasion on which Napoleon tried to get a rise out of Laplace by protesting that he had written a huge book on the system of the world without once mentioning God as the author of the universe. Laplace is supposed to have replied, "Sire, I had no need of that hypothesis." The principal legacy of the *Mécanique Céleste* to later generations lay in Laplace's wholesale development of potential theory, with its far-reaching implications for a dozen different branches of physical science ranging from gravitation and fluid mechanics to electro-magnetism and atomic physics. Even though he lifted the idea of the potential from Lagrange without acknowledgment, he exploited it so extensively that ever since his time the fundamental differential equation of potential theory has been known as Laplace's equation.

His other masterpiece was the treatise *Théorie Analytique des Probabilités* (1812), in which he incorporated his own discoveries in probability from the preceding 40 years. Again he failed to acknowledge the many ideas of others he mixed in with his own; but even discounting this, his book is generally agreed to be the greatest contribution to this part of mathematics by any one man. In the introduction he says: "At bottom, the theory of probability is only common sense reduced to calculation." This may be so, but the following 700 pages of intricate analysis—in which he freely used Laplace transforms, generating functions, and many other highly nontrivial tools—has been said by some to surpass in complexity even the *Mécanique Céleste*.

After the French Revolution Laplace's political talents and greed for position came to full flower. His countrymen speak ironically of his "suppleness" and "versatility" as a politician. What this really means is that each time there was a change of regime (and there were many), Laplace smoothly adapted himself by changing his principles—back and forth between fervent republicanism and fawning royalism—and each time he emerged with a better job and grander titles. He has been aptly compared with the apocryphal Vicar of Bray in English literature, who was twice a Catholic and twice a Protestant. The Vicar is said to have replied as follows to the charge of being a turncoat: "Not so, neither, for if I changed my religion, I am sure I kept true to my principle, which is to live and die the Vicar of Bray."

To balance his faults, Laplace was always generous in giving assistance and encouragement to younger scientists. From time to time he helped forward in their careers such men as the chemist Gay-Lussac, the traveler and naturalist Humboldt, the physicist Poisson, and—appropriately—the young Cauchy, who was destined to become one of the chief architects of nineteenth century mathematics.

Appendix B. Abel

Niels Henrik Abel (1802–1829) was one of the foremost mathematicians of the nineteenth century and probably the greatest genius produced by the Scandinavian countries. Along with his contemporaries Gauss and Cauchy, Abel was one of the pioneers in the development of modern mathematics, which is characterized by its insistence on rigorous proof. His career was a poignant blend of good-humored optimism under the strains of poverty and neglect, modest satisfaction in the many towering achievements of his brief maturity, and patient resignation in the face of an early death.

Abel was one of six children in the family of a poor Norwegian country minister. His great abilities were recognized and encouraged by one of his teachers when he was only sixteen, and soon he was reading and digesting the works of Newton, Euler, and Lagrange. As a comment on this experience, he inserted the following marginal remark in one of his later mathematical notebooks: "It appears to me that if one wants to make progress in mathematics, one should study the masters and not the pupils." When Abel was only eighteen his father died and left the family destitute. They subsisted by the aid of friends and neighbors, and somehow the boy, helped by contributions from several professors, managed to enter the University of Oslo in 1821. His earliest researches were published in 1823, and included his solution of the classic tautochrone problem by means of the integral equation discussed in [Section 52](#). This was the first solution of an equation of this kind, and foreshadowed the extensive development of integral equations in

the late nineteenth and early twentieth centuries. He also proved that the general fifth degree equation $ax^5 + bx^4 + cx^3 + dx^2 + ex + f = 0$ cannot be solved in terms of radicals, as is possible for equations of lower degree, and thus disposed of a problem that had baffled mathematicians for 300 years. He published his proof in a small pamphlet at his own expense.

In his scientific development Abel soon outgrew Norway, and longed to visit France and Germany. With the backing of his friends and professors he applied to the government, and after the usual red tape and delays, he received a fellowship for a mathematical grand tour of the Continent. He spent most of his first year abroad in Berlin. Here he had the great good fortune to make the acquaintance of August Leopold Crelle, an enthusiastic mathematical amateur who became his close friend, advisor, and protector. In turn, Abel inspired Crelle to launch his famous *Journal für die Reine und Angewandte Mathematik*, which was the world's first periodical devoted wholly to mathematical research. The first three volumes contained 22 contributions by Abel.

Abel's early mathematical training had been exclusively in the older formal tradition of the eighteenth century, as typified by Euler. In Berlin he came under the influence of the new school of thought led by Gauss and Cauchy, which emphasized rigorous deduction as opposed to formal calculation. Except for Gauss's great work on the hypergeometric series, there were hardly any proofs in analysis that would be accepted as valid today. As Abel expressed it in a letter to a friend: "If you disregard the very simplest cases, there is in all of mathematics not a single infinite series whose sum has been rigorously determined. In other words, the most important parts of mathematics stand without a foundation." In this period he wrote his classic study of the binomial series, in which he founded the general theory of convergence and gave the first satisfactory proof of the validity of this series expansion.

Abel had sent to Gauss in Göttingen his pamphlet on the fifth degree equation, hoping that it would serve as a kind of scientific passport. However, for some reason Gauss put it aside without looking at it, for it was found uncut among his papers after his death 30 years later. Unfortunately for both men, Abel felt that he had been snubbed, and decided to go on to Paris without visiting Gauss.

In Paris he met Cauchy, Legendre, Dirichlet, and others, but these meetings were perfunctory and he was not recognized for what he was. He had already published a number of important articles in Crelle's *Journal*, but the French were hardly aware yet of the existence of this new periodical and Abel was much too shy to speak of his own work to people he scarcely knew. Soon after his arrival he finished his great *Mémoire sur une Propriété Générale d'une Classe Très Étendue des Fonctions Transcendantes*, which he regarded as his masterpiece. This work contains the discovery about integrals of algebraic functions now known as Abel's theorem, and is the foundation for the later theory of Abelian integrals, Abelian functions, and much of algebraic

geometry. Decades later, Hermite is said to have remarked of this *Mémoire*: “Abel has left mathematicians enough to keep them busy for 500 years.” Jacobi described Abel’s theorem as the greatest discovery in integral calculus of the nineteenth century. Abel submitted his manuscript to the French Academy. He hoped that it would bring him to the notice of the French mathematicians, but he waited in vain until his purse was empty and he was forced to return to Berlin. What happened was this: the manuscript was given to Cauchy and Legendre for examination; Cauchy took it home, mislaid it, and forgot all about it; and it was not published until 1841, when again the manuscript was lost before the proof sheets were read. The original finally turned up in Florence in 1952.⁵ In Berlin, Abel finished his first revolutionary article on elliptic functions, a subject he had been working on for several years, and then went back to Norway, deeply in debt.

He had expected on his return to be appointed to a professorship at the university, but once again his hopes were dashed. He lived by tutoring, and for a brief time held a substitute teaching position. During this period he worked incessantly, mainly on the theory of the elliptic functions that he had discovered as the inverses of elliptic integrals. This theory quickly took its place as one of the major fields of nineteenth century analysis, with many applications to number theory, mathematical physics, and algebraic geometry. Meanwhile, Abel’s fame had spread to all the mathematical centers of Europe and he stood among the elite of the world’s mathematicians, but in his isolation he was unaware of it. By early 1829 the tuberculosis he contracted on his journey had progressed to the point where he was unable to work, and in the spring of that year he died, at the age of twenty-six. As an ironic postscript, shortly after his death Crelle wrote that his efforts had been successful, and that Abel would be appointed to the chair of mathematics in Berlin.

Crelle eulogized Abel in his *Journal* as follows: “All of Abel’s works carry the imprint of an ingenuity and force of thought which is amazing. One may say that he was able to penetrate all obstacles down to the very foundation of the problem, with a force which appeared irresistible... He distinguished himself equally by the purity and nobility of his character and by a rare modesty which made his person cherished to the same unusual degree as was his genius.” Mathematicians, however, have their own ways of remembering their great men, and so we speak of Abel’s integral equation, Abelian integrals and functions, Abelian groups, Abel’s series, Abel’s partial summation formula, Abel’s limit theorem in the theory of power series, and Abel summability. Few have had their names linked to so many concepts and theorems in modern mathematics, and what he might have accomplished in a normal lifetime is beyond conjecture.

⁵ For the details of this astonishing story, see the fine book by O. Ore, *Niels Henrik Abel: Mathematician Extraordinary*, University of Minnesota Press, Minneapolis, 1957.